A Scaled and Adaptive FISTA Algorithm for Signal-Dependent Sparse Image Super-Resolution Problems

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Abstract. We propose a scaled adaptive version of the Fast Iterative Soft-Thresholding Algorithm, named S-FISTA, for the efficient solution of convex optimization problems with sparsity-enforcing regularization. S-FISTA couples a non-monotone backtracking procedure with a scaling strategy for the proximal–gradient step, which is particularly effective in situations where signal-dependent noise is present in the data. The proposed algorithm is tested on some image super-resolution problems where a sparsity-promoting regularization term is coupled with a weighted-ℓ_2 data fidelity. Our numerical experiments show that S-FISTA allows for faster convergence in function values with respect to standard FISTA, as well as being an efficient inner solver for iteratively reweighted ℓ_1 algorithms, thus reducing the overall computational times.

Keywords: Inertial forward-backward splitting · Variable metric · Sparse super-resolution · Scaled FISTA · Sparse optimization

1 Introduction

Among the plethora of first-order convex optimization schemes proposed over the last decades to solve image reconstruction problems (see [6] for a review), forward-backward (FB) algorithms have attracted a significant attention due to their easy applicability in many scenarios and their fast convergence properties whenever endowed with appropriate inertial updates, as in the case of the celebrated FISTA algorithm [1]. Their successful application relies, in practice, on the accurate estimation of the 'steepness' of the smooth component of the functional to minimize; whenever such estimate is coarse, the practical effectiveness of FB methods may be limited. One remedy to solve this issue consists in the

© Springer Nature Switzerland AG 2021
A. Elmoataz et al. (Eds.): SSVM 2021, LNCS 12679, pp. 242–253, 2021.
https://doi.org/10.1007/978-3-030-75549-2_20

adoption of adaptive backtracking procedures, see e.g. [5,16]. Moreover, as noted in several works in the literature [3,7,10], FB algorithms can also benefit from suitable scaling approaches aimed at capturing some second-order information of the smooth component. In particular, the use of scaling approaches has been shown to render particularly effective in the context of signal-dependent image reconstruction problems in a variety of applications ranging from biological to astronomical imaging, see e.g., [2]. Recently, in [14] the authors proposed SAGE-FISTA, a general FISTA-type algorithm designed for (possibly strongly) convex composite problems where a suitable scaling strategy is combined with an adaptive backtracking technique. Upon suitable conditions on the scaling updates, accelerated convergence rates are rigorously proved for SAGE-FISTA.

In this work we consider a particular instance of the SAGE-FISTA algorithm (denominated S-FISTA) for efficiently solving sparse reconstruction problems arising in the field of image super-resolution with signal-dependent Poisson noise. The use of composite functionals promoting signal sparsity by means of both convex and non-convex regularizers has been extensively explored in recent years, see, e.g. [18]. However, as preliminary results showed in [11], when such models are used in correspondence with data terms explicitly incorporating signal-dependence, the numerical efficiency of standard FISTA-type algorithms may suffer due to the use of excessively small step-sizes. By contrast, our numerical experience shows that S-FISTA significantly improves computational convergence when used both as global solver in convex optimization regimes and as inner solver for iterative non-convex optimization schemes such as iteratively reweighted ℓ_1 algorithm [13]. The main contribution of this work is the use of a tailored efficient algorithm for improving overall computational efficiency.

2 Scaled and Adaptive FISTA

We consider the following convex composite problem

$$\min_{\mathbf{x}\in\mathbb{R}^N} F(\mathbf{x}) := f(\mathbf{x}) + g(\mathbf{x}), \tag{1}$$

where $f : Y \to \mathbb{R}$ is a convex, continuously differentiable function with $\mathcal{L}-$Lipschitz continuous gradient on a closed, convex set $Y \subseteq \mathbb{R}^N$ and $g : \mathbb{R}^N \to \bar{\mathbb{R}}$ is convex and possibly non-smooth.

Forward–backward (FB) algorithms are classical optimization tools for solving (1), whose main iteration consists in alternating a gradient step on the differentiable part f with a proximal step on the term g [8]. The following sequence of iterates is thus generated:

$$\begin{aligned}\mathbf{x}^{k+1} &= \mathrm{prox}_{\tau_{k+1}g}(\mathbf{x}^k - \tau_{k+1}\nabla f(\mathbf{x}^k)) \\ &= \underset{\mathbf{z}\in\mathbb{R}^N}{\mathrm{argmin}}\ g(\mathbf{z}) + \frac{1}{2\tau_{k+1}}\|\mathbf{z} - \mathbf{x}^k + \tau_{k+1}\nabla f(\mathbf{x}^k)\|^2, \quad k \geq 0,\end{aligned} \tag{2}$$

where $\tau_{k+1} > 0$ is the step-size parameter. The standard scheme (2) is usually accelerated by introducing an inertial term in the proximal–gradient step, as in

the popular Fast Iterative Soft-Thresholding Algorithm (FISTA) [1], where each iterate is obtained by extrapolating the two previous iterates as follows:

$$\mathbf{y}^{k+1} = \mathbf{x}^k + \left(\frac{t_k - 1}{t_{k+1}}\right)(\mathbf{x}^k - \mathbf{x}^{k-1}), \quad \mathbf{x}^{k+1} = \mathrm{prox}_{\tau_{k+1}g}(\mathbf{y}^{k+1} - \tau_{k+1}\nabla f(\mathbf{y}^{k+1})),$$

where $t_0 = 1$ and $t_{k+1} = \frac{1+\sqrt{1+4t_k^2}}{2}$. FISTA enjoys a favorable $\mathcal{O}(1/k^2)$ convergence rate, which is optimal for first-order methods, see [1,12]. However, it is also well-known that the practical performance of FISTA may not be as satisfactory as its theoretical convergence rate suggests. This is mainly due to the restricted range of admissible step-sizes τ_{k+1} discussed in [1] to ensure convergence, which requires computing τ_{k+1} either as $1/\mathcal{L}$ for all $k \geq 0$ or using a backtracking procedure that monotonically reduces τ_{k+1} until an estimate of $\mathcal{L}$ is found. Such an approach may negatively affect the practical convergence rate of FISTA, especially when only a large (pessimistic) estimate of $\mathcal{L}$ is known.

In [14] the authors propose the so-called Scaled Adaptive GEneralized FISTA (SAGE-FISTA), a novel variant of FISTA aimed at improving its practical convergence rate. SAGE-FISTA employs an adaptive backtracking strategy allowing the adaptive (increasing/decreasing) adjustment of τ_{k+1} at each iteration, which comes in handy when the initial τ_0 is extremely small due to a large estimate of $\mathcal{L}$. The adaptive backtracking is further combined with a symmetric and positive definite scaling matrix $\mathbf{D}_{k+1}$ in the proximal–gradient step, by which some second-order information of the function f around the current iterate $\mathbf{x}^k$ may be incorporated while keeping the iteration computationally inexpensive, e.g. adopting Majorization-Minimization techniques or split-gradient strategies [4,7]. The algorithm in [14] also takes into account the inexact computation of the proximal

Algorithm 1. S-FISTA($\mathbf{x}^0$,τ_0,f,g)

Parameters: $\rho \in (0,1)$, $\delta \in (0,1]$, $\{\eta_{inf}^k\}_k$, $\{\eta_{sup}^k\}_k$ s.t. $0 < \eta_{inf} \leq \eta_{inf}^k \leq \eta_{sup}^k \leq \eta_{sup}$.

Initialization: $\mathbf{x}^{-1} = \mathbf{x}^0$, $t_0 \geq 1$, $\mathbf{D}_0 \in \mathcal{D}_{\eta_{inf}^0}^{\eta_{sup}^0}$.

FOR $k = 0, 1, \ldots$ REPEAT

Choose $\mathbf{D}_{k+1} \in \mathcal{D}_{\eta_{inf}^k}^{\eta_{sup}^k}$ and set $\tau_{k+1}^0 = \frac{\tau_k}{\delta}$.

FOR $i = 0, 1, \ldots$ REPEAT

STEP 1. Set $\tau_{k+1} = \rho^i \tau_{k+1}^0$.

STEP 2. Set $t_{k+1} = \frac{1+\sqrt{1+4\frac{\tau_k}{\tau_{k+1}}t_k^2}}{2}$.

STEP 3. Set $\mathbf{y}^{k+1} = \mathrm{proj}_{Y,\mathbf{D}_{k+1}}\left(\mathbf{x}^k + \left(\frac{t_k-1}{t_{k+1}}\right)(\mathbf{x}^k - \mathbf{x}^{k-1})\right)$.

STEP 4. Compute the next iterate as

$$\mathbf{x^{k+1}} = \mathrm{prox}_{\tau_{k+1}g}^{\mathbf{D}_{k+1}}(\mathbf{y}^{k+1} - \tau_{k+1}\mathbf{D}_{k+1}^{-1}\nabla f(\mathbf{y}^{k+1})).$$

UNTIL $f(\mathbf{x}^{k+1}) \leq f(\mathbf{y}^{k+1}) + \nabla f(\mathbf{y}^k)^T(\mathbf{x}^{k+1} - \mathbf{y}^{k+1}) + \frac{1}{2\tau_{k+1}}\|\mathbf{x}^{k+1} - \mathbf{y}^{k+1}\|_{\mathbf{D}_{k+1}}^2$

UNTIL stopping criterion

operator and the possible introduction of the strong convexity moduli of either f or g. Here, we consider a simplified, convex version, denominated Scaled FISTA (S-FISTA), where the proximal operator of g is assumed to be computable in closed form. We report S-FISTA in Algorithm 1.

Let $\mathcal{SP}(\mathbb{R}^{N\times N})$ denote the set of $N \times N$ symmetric, positive definite real matrices and $\mathcal{D}_{\eta^k_{inf}}^{\eta^k_{sup}} = \{\mathbf{D} \in \mathcal{SP}(\mathbb{R}^{N\times N}) : \ \eta^k_{inf}\|x\|^2 \leq \|x\|^2_{\mathbf{D}} \leq \eta^k_{sup}\|x\|^2, \ \forall\ x\}$, where $\|x\|_{\mathbf{D}} = \sqrt{x^T\mathbf{D}x}$ denotes the norm induced by the matrix $\mathbf{D}$.

For all iterations $k \geq 0$, we first select a scaling matrix $\mathbf{D}_{k+1} \in \mathcal{D}_{\eta^k_{inf}}^{\eta^k_{sup}}$ and a tentative step-size $\tau^0_{k+1} = \tau_k/\delta$. Note that if $\delta < 1$, this corresponds to an increasing of τ_k, whereas when $\delta = 1$, nothing happens at this step. Then, the computation of $\mathbf{x}^{k+1}$ is performed by means of an inner backtracking procedure depending on the scaling matrix $\mathbf{D}_{k+1}$. At each inner backtracking iteration $i \geq 0$, we set the parameter t_{k+1} and the projected inertial point $\mathbf{y}^{k+1}$, where $\mathrm{proj}_{Y,\mathbf{D}_{k+1}}(\mathbf{x}) = \mathrm{argmin}_{\mathbf{z}\in Y}\ \|\mathbf{z}-\mathbf{x}\|^2_{\mathbf{D}_{k+1}}$ denotes the projection operator onto Y w.r.t. the norm induced by $\mathbf{D}_{k+1}$. The proximal–gradient point $\mathbf{x}^{k+1}$ is then computed by pre-multiplying $\nabla f(\mathbf{y}^{k+1})$ with $\mathbf{D}^{-1}_{k+1}$ and performing a scaled proximal step, where $\mathrm{prox}^{\mathbf{D}_{k+1}}_{\tau_{k+1}g}(\mathbf{x}) = \mathrm{argmin}_{\mathbf{z}\in\mathbb{R}^N}\ \ g(\mathbf{z}) + \frac{1}{2}\|\mathbf{z}-\mathbf{x}\|^2_{\mathbf{D}_{k+1}}$ is the proximal operator in the norm induced by $\mathbf{D}_{k+1}$. Finally, a check is made on a sufficient decrease condition based on a scaled version of the classical descent lemma [1, Lemma 2.1]; if such a condition is not met, the step-size is reduced by a factor $\rho \in (0,1)$ and the backtracking steps are performed again until the condition holds. Note that this procedure ends in a finite number of steps, since the decrease condition is verified for sufficiently small step-sizes [14, Lemma 2.2].

If the bounds $\eta^k_{inf}, \eta^k_{sup}$ converge sufficiently fast to the same constant value, then it is possible to prove an optimal $\mathcal{O}(1/k^2)$ convergence rate for S-FISTA.

Theorem 1. *[14, Corollary 3.3] Let $\mathbf{x}^*$ be the solution of problem (1) and suppose the parameters $\eta^k_{inf}, \eta^k_{sup}$ are computed as*

$$\eta^k_{inf} = \eta - \nu^k_{inf}, \quad \eta^k_{sup} = \eta + \nu^k_{sup}, \tag{3}$$

where $\nu^k_{inf}, \nu^k_{sup} \geq 0$ and $\sum_{k=0}^{\infty}\nu^k_{inf} < \infty$, $\sum_{k=0}^{\infty}\nu^k_{sup} < \infty$. Then, for all $k \geq 0$

$$F(\mathbf{x}^{k+1}) - F(\mathbf{x}^*) \leq \frac{C}{(k+1+t_0)^2}, \tag{4}$$

where $C = C(\mathbf{x}^*, \mathbf{x}^0, \tau_0, \mathbf{D}_0, t_0, \rho, \eta_{inf}, \eta_{sup}) > 0$ is a constant depending on $\mathbf{x}^*$ and the parameters of Algorithm 1.

We refer the reader to Sect. 4 for a practical implementation of (3).

S-FISTA has already proven to be effective on several imaging problems (see [14, Section 4]) thanks to the combination of the scaling procedure with the adaptive backtracking, which allows to select bigger step-sizes, thus improving the practical convergence speed towards the solution of the problem. We now show the effectiveness of S-FISTA on image super-resolution problems.

3 Application to Sparse Weighted Models for Super-Resolution Microscopy

In several works (see, e.g., [3,10]) it has been shown that variable metric first-order methods perform particularly well in image reconstruction/restoration problems where data are corrupted by signal-dependent Poisson noise, as it is common, for instance, in the field of image microscopy. In the following, we focus on the problem of image super-resolution.

3.1 Sparse Image Super-Resolution with Poisson Data

The goal of super-resolution fluorescence image microscopy is to estimate molecules' intensities and positions from blurred and noisy acquisitions of sparse samples of molecules. Namely, for a given Low Resolution (LR), blurred and noisy (vectorized) image $\mathbf{y} \in \mathbb{R}^M_{>0}$, the ill-posed inverse problem of reconstructing a Super-Resolved image $\mathbf{x} \in \mathbb{R}^N_{\geq 0}$, where $N = L^2 M$ and $L > 0$ denotes the Super-Resolution (SR) factor, consists in solving:

$$\text{find} \quad \mathbf{x} \quad \text{s.t.} \quad \mathbf{y} = \mathcal{P}(\mathbf{R_L H x}), \tag{5}$$

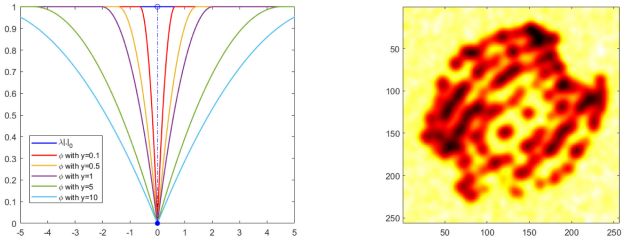
where $\mathbf{H} \in \mathbb{R}^{N \times N}$ is the blurring operator corresponding to the microscope PSF $\mathbf{h}$, $\mathbf{R_L} \in \mathbb{R}^{M \times N}$ is the under-sampling operator and $\mathcal{P}(\mathbf{z})$ denotes the realization of a Poisson-distributed random vector with mean values $(\mathbf{z})_i \geq 0$, for $i = 1, \ldots, M$ which are assumed to be strictly positive for simplicity. To simplify notations, we will denote in the following by $\mathbf{A} := \mathbf{R_L H} \in \mathbb{R}^{M \times N}$ the whole forward operator. Following a standard Maximum A Posteriori estimation yields to having the Kullback-Leibler (KL) divergence $D_{KL}(\mathbf{Ax}; \mathbf{y}) := \sum_{j=1}^M (\mathbf{Ax})_j - y_j \log(\mathbf{Ax})_j$ as data fidelity term. Several approximations of D_{KL} have been considered in the literature (see [15]); in the following, we consider the second order Taylor expansion around $\mathbf{y}$ which results in a weighted-ℓ_2 fidelity term:

$$\frac{1}{2} \sum_{i=1}^M \frac{((\mathbf{Ax})_i - y_i)^2}{y_i} = \frac{1}{2} \langle \mathbf{Ax} - \mathbf{y}, \mathbf{W}(\mathbf{Ax} - \mathbf{y}) \rangle =: \frac{1}{2} \|\mathbf{Ax} - \mathbf{y}\|^2_{\mathbf{W}}, \tag{6}$$

where the weighted norm is defined in terms of the diagonal positive definite matrix $\mathbf{W} := \text{diag}(\mathbf{1}/\mathbf{y}) \in \mathbb{R}^{M \times M}$ where division is intended element-wise. Upon this choice and consistently with the signal-dependent modeling considered, the weighted-ℓ_2 norm measures the relative discrepancy between the underlying noiseless signal $\mathbf{Ax}$ and the noisy signal $\mathbf{y}$ component-wise, so as to have a low data-fidelity whenever intense signal (hence high noise) is measured.

The natural choice for promoting sparsity in the desired signal consists in considering the ℓ_0 pseudo-norm or its continuous relaxations, see, e.g. [18]. Coupling such regularization with the fidelity term (6), we can thus consider the following minimization problem:

$$\hat{\mathbf{x}} \in \arg\min_{\mathbf{x} \in \mathbb{R}^N} \; G_{w\ell_0}(\mathbf{x}) \equiv \frac{1}{2} \|\mathbf{Ax} - \mathbf{y}\|^2_{\mathbf{W}} + \lambda \|\mathbf{x}\|_0 + i_{\geq 0}(\mathbf{x}), \tag{7}$$



(a) $\phi(\frac{a}{\sqrt{y}}, \lambda, u)$ (b) $\|\mathbf{a_i}\|_{\mathbf{W}}$

Fig. 1. Left: Dependence of the penalty in (9) on the data y for fixed $\lambda = 1$ and $a = 0.7$. Right: plot of parameters used for the test image shown in Fig. 2(b).

where $\lambda > 0$ is the regularization parameter and $i_{\geq 0}(\cdot)$ denotes the indicator function of $Y := \{x \in \mathbb{R}^N | \ x_i \geq 0 \ \forall i = 1, \ldots, N\}$ which forces the desired solution $\mathbf{x}$ to be non-negative. Note that the functional $G_{w\ell_0}$ is not continuous nor convex, due to the presence of the ℓ_0 pseudo-norm which further makes the problem NP-hard. A common strategy to mitigate such numerical difficulties is to consider a continuous relaxation of the ℓ_0 pseudo-norm, in order to make use of standard optimization algorithms, such as, for instance, the Iterative Reweighted ℓ_1 (IRL1) algorithm [13]. We consider here the weighted-Continuous Exact ℓ_0 (wCEL0) relaxation recently proposed in [11]. Inspired by [17], the relaxed model computed by double Fenchel biconjugation applied to $G_{w\ell_0}$ reads:

$$\hat{\mathbf{x}} \in \arg\min_{\mathbf{x} \in \mathbb{R}^N} \ G_{\texttt{wCEL0}}(\mathbf{x}) \equiv \frac{1}{2}||\mathbf{A}\mathbf{x} - \mathbf{y}||^2_{\mathbf{W}} + \Phi_{\texttt{wCEL0}}(\mathbf{x}; \lambda, \mathbf{A}, \mathbf{W}) + \iota_{\geq 0}(\mathbf{x}), \tag{8}$$

where the continuous and non-convex functional $\Phi_{\texttt{wCEL0}}(\cdot; \lambda, \mathbf{A}, \mathbf{W})$ is defined by

$$\Phi_{\texttt{wCEL0}}(\mathbf{x}; \lambda, \mathbf{A}, \mathbf{W}) = \sum_{i=1}^{N} \phi(\|\mathbf{a}_i\|_{\mathbf{W}}, \lambda, |x_i|), \tag{9}$$

for $\lambda > 0$, with $\mathbf{a_i} \in \mathbb{R}^M$ being i-th column of the matrix $\mathbf{A}$ for $i = 1, \ldots, N$ and

$$\phi(\bar{a}, \lambda, u) = \lambda - \frac{\bar{a}^2}{2}\left(u - \frac{\sqrt{2\lambda}}{\bar{a}}\right)^2 \mathbb{1}_{\{u < \frac{\sqrt{2\lambda}}{\bar{a}}\}}. \tag{10}$$

We remark that (9) depends both on the forward model and on the observed data and it is a sum of N one-dimensional functions. The expression of (10) is obtained by a direct analytical computation of the convex envelope of the corresponding 1D restriction of the $G_{w\ell_0}$ functional (7) and $\bar{a}$ encodes the data-dependence of the 1D penalty as $\bar{a} = \frac{a}{\sqrt{y}}$, with $a > 0$ being the 1D forward model and $y > 0$ the observed data with background. In $\Phi_{\texttt{wCEL0}}$, the dependence on the data is reflected in the weighted column norms $\bar{a} = \|\mathbf{a}_i\|_{\mathbf{W}}$, which depend on the data $\mathbf{y}$, as shown in Fig. 1.

Compared to the Continuous Exact ℓ_0 (CEL0) $\Phi_{\texttt{CEL0}}$ penalty studied in [17], the $\Phi_{\texttt{WCEL0}}$ penalty is more suited to deal with signal-dependent noise models. Moreover, the data term (6) offers the opportunity to improve the computational performance of optimization solvers by taking advantage of scaling techniques.

3.2 Solving (8) Efficiently: IRL1 Algorithm with S-FISTA Solver

The composite non-convex continuous problem (8) can be solved by means of an iteratively reweighted ℓ_1 (IRL1) scheme [13] combined with S-FISTA Algorithm 1 as inner solver. Iteratively reweigthed schemes are based on the successive minimization of a sequence of convex weighted ℓ_1 subproblems, each obtained by appropriately linearizing the (possibly non-convex) penalty term. In the case of (8), following [13, Algorithm 3], one can replace each term $\phi(\|\mathbf{a}_i\|_{\mathbf{W}}, \lambda, |x_i|)$ in (9) with the linear function $x_i \mapsto \omega_i^{\mathbf{x}^k}|x_i|$, where

$$\omega_i^{\mathbf{x}^k} = \left(\sqrt{2\lambda} - \|\mathbf{a}_i\|_{\mathbf{W}}^2 |x_i^k|\right) \mathbb{1}_{\{|x_i^k| < \frac{\sqrt{2\lambda}}{\|\mathbf{a}_i\|_{\mathbf{W}}}\}} \in \partial_{x_i}\phi(\|\mathbf{a}_i\|_{\mathbf{W}}, \lambda, |x_i|^k)$$

is the generalized derivative of $\phi(\|\mathbf{a}_i\|_{\mathbf{W}}, \lambda, \cdot)$ at point $|x_i|^k$. This generates the following sequence of convex subproblems:

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}\in\mathbb{R}^N} \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 + \lambda \sum_{i=1}^{N} \omega_i^{\mathbf{x}^k}|x_i| + \iota_{\geq 0}(\mathbf{x}), \quad k \geq 0, \tag{11}$$

which do not usually admit a closed-form solution and hence require to be solved approximately via an iterative optimization solver able to deal with weighted ℓ_1 terms, such as ISTA or FISTA. As highlighted in Sect. 2, these solvers may be extremely inefficient when only a pessimistic estimate of the Lipschitz constant $\mathcal{L}$ of the gradient is at our disposal. In particular, this is true for problem (11) for which, by denoting by $\mathcal{F}(\cdot)$ the 2D Fourier transform operator, we get

$$\mathcal{L} = \|\mathbf{A}^T\mathbf{W}\mathbf{A}\|_2 \leq \|\mathbf{A}\|^2\|\mathbf{W}\| = L^2 \max(\mathcal{F}(\mathbf{h}))^2 \min(\mathbf{y})^{-1}, \tag{12}$$

which may be extremely large due to the possible values close to zero assumed by $\mathbf{y}$. Such a large estimate leads to a sequence a small step-sizes slowing down the convergence speed of the inner solver, which in turn worsen the progress of the outer IRL1 scheme towards a stationary point of (8).

In order to solve problem (8) more efficiently, we propose to employ the S-FISTA Algorithm 1 for computing inexactly the IRL1 iterate (11). The resulting scheme is reported in Algorithm 2.

Note that a warm-start strategy is adopted inside Algorithm 2, as the previous outer iterate $\mathbf{x}^k$ and step-size τ_k are used as initial guesses for the S-FISTA inner solver. We also remark that the convergence of the IRL1 scheme to a stationary point of (8) has been proved so far only when the subproblems (11) are solved exactly (see [13,17]), whereas the convergence of Algorithm 2 still needs to be addressed. Nonetheless, the numerical performance of Algorithm 2 is promising, as we show in the next section.

4 Numerical Results

In this section we report two tests confirming the effectiveness of S-FISTA when used both in convex optimization regimes as a global solver (Algorithm 1) and in non-convex scenarios as an efficient inner routine for the IRL1 in Algorithm 2. In both cases, the weighted-ℓ_2 data term (6) is used as the smooth component f in the splitting (1). Concerning the choice of the scaling matrices $\{\mathbf{D}_k\}_k$, we consider the split-gradient strategy proposed in [10] where the decomposition $-\nabla f(\mathbf{x}) = U(\mathbf{x}) - V(\mathbf{x})$ with $U(\mathbf{x}) \geq 0$ and $V(\mathbf{x}) > 0$ and the choice $\mathbf{D}_k = \operatorname{diag}\left(\mathbf{y}^{(k)}/V(\mathbf{y}^{(k)})\right)^{-1}$ is made. In order to ensure condition (3), we further introduce an appropriate sequence of thresholding parameters $\{\gamma_k\}_k$ in the definition of $\mathbf{D}_k$, thus obtaining

Algorithm 2. Inexact IRL1 algorithm for Weighted CEL0

Input: $\mathbf{y} \in \mathbb{R}^M$, $\mathbf{x}^0 \in \mathbb{R}^N$, $\tau_0 > 0$, $\lambda > 0$
FOR $k = 0, 1, \ldots$ REPEAT

STEP 1. Update the weights $\omega_i^{\mathbf{x}^k} \in \partial_{x_i}\phi(\|a_i\|_{\mathbf{W}}, \lambda, |x_i|^k)$.
STEP 2. Set $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2$, $g^k(\mathbf{x}) := \lambda \sum_{i=1}^N \omega_i^{\mathbf{x}^k}|x_i| + \iota_{\geq 0}(\mathbf{x})$ and compute

$$\mathbf{x}^{k+1} = \text{S-FISTA}(\mathbf{x}^k, \tau_k, f, g^k).$$

UNTIL stopping criterion
Output: $\mathbf{x}$

$$\mathbf{D}_k = \operatorname{diag}\left(\max\left(\frac{1}{\gamma_k}, \min\left(\gamma_k, \frac{\mathbf{y}^{(k)}}{V(\mathbf{y}^{(k)})}\right)\right)\right)^{-1}, \quad \gamma_k = \sqrt{1 + \frac{s_1}{(k+1)^{s_2}}}, \tag{13}$$

where $s_1 > 0$, $s_2 > 0$. Note that when $s_1 = 0$, then $\mathbf{D}_k \equiv \mathbf{I}$, hence the standard Euclidean metric is recovered. In general, large values of s_1 promote the use of the variable metric in the early iterations, while the behavior for $k \gg 1$ is driven by s_2 which determines the speed of convergence of $\mathbf{D}_k$ towards $\mathbf{I}$. For the term (6), simple calculations show that $V(\mathbf{y}^{(k)}) = \tilde{\mathbf{A}}^T\tilde{\mathbf{A}}\mathbf{y}^{(k)}$, with $\tilde{\mathbf{A}} := \sqrt{\mathbf{W}}\mathbf{A}$.

4.1 Weighted-ℓ_2-ℓ_1 Minimization

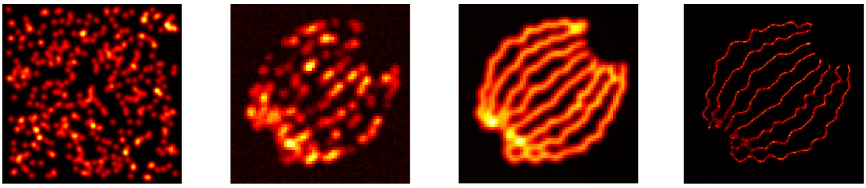
As a preliminary test, we consider the S-FISTA Algorithm 1 endowed with both Armijo ($\delta = 1$) and adaptive ($\delta < 1$) backtracking to solve the weighted-$\ell_2 - \ell_1$ convex optimization problem:

$$\hat{\mathbf{x}} \in \arg\min_{\mathbf{x}\in\mathbb{R}^N} \; F(\mathbf{x}) \equiv \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_{\mathbf{W}}^2 + \lambda\|\mathbf{x}\|_1 + \iota_{\geq 0}(\mathbf{x}). \tag{14}$$

For this test we considered fluorescent microscopy data $\mathbf{y} \in \mathbb{R}_{>0}^M$ with $M = 128^2$ simulated by generating a random density of points uniformly distributed over

the image domain $\mathbf{\Omega}$, then corrupted by Gaussian blur with standard deviation $\sigma_{\mathbf{h}} = 150$ nm and Poisson noise by using the inbuilt `imnoise` MATLAB routine, see Fig. 2(a). As SR factor, we set $L = 4$.

S-FISTA Algorithm 1 is applied to solve problem (14) with the following parameters: $\lambda = 0.85$, $\mathbf{x}^0 = \mathbf{A}^T\mathbf{y}$ maximum number of outer iterations `maxiter` $= 500$, maximum number of backtracking iterations `max_bt` $= 10$, $\rho = 0.8$ (decreasing backtracking factor), $L_0 = 1/\tau_0 = 100$ and $t_0 = 1.01$. In order to compare the monotone backtracking strategy with the adaptive one, we then choose $\delta = 0.99$ and $\delta = 1$, respectively. Note that STEP 4 can be here computed in closed-form by rescaled soft-thresholding.



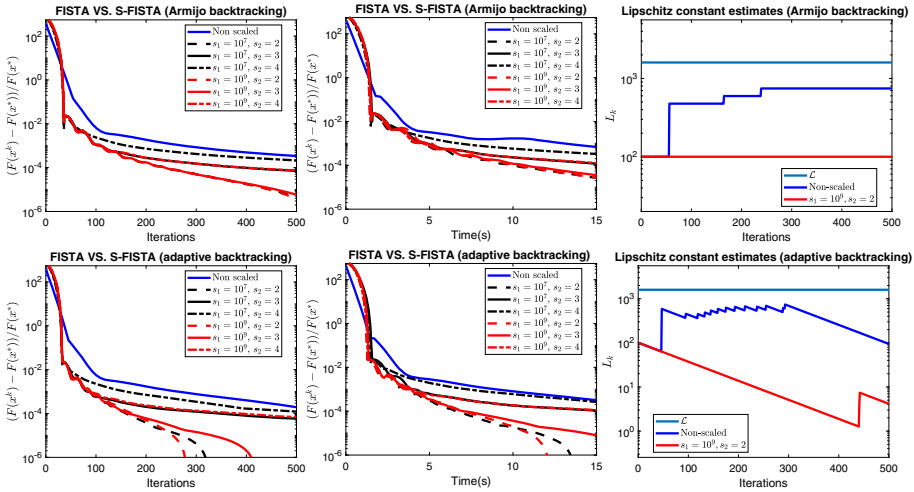
(a) `density` (b) ISBI 2013, single frame (c) ISBI 2013, $\bar{\mathbf{y}}$ (d) IRL1+S-FISTA adapt.

Fig. 2. Test images used for experiments and results obtained by applying the IRL1+S-FISTA Algorithm 1 on the whole ISBI 2013 image dataset.

In Figs. 3 we compare the performance of standard (i.e. non-scaled) FISTA with its scaled variant S-FISTA used for solving (14) for different choices of the scaling matrices $\{\mathbf{D}_k\}_k$ in (13) and for monotone (first row) and adaptive (second row) backtracking. To do so, we compute the relative error $(F(\mathbf{x}^k) - F(\mathbf{x}^*))/F(\mathbf{x}^*)$ (where $\mathbf{x}^*$ has been pre-computed) calculated both along iterations (first column) and for a maximum run of 15 s (second column) of CPU time. In the third column we report the values of the estimated Lipschitz constant L_k along iterations. We observe that a *sage* combination of scaling (for selected values of s_1 and s_2 in (13)) and adaptive backtracking significantly improves convergence speed and computational performance.

4.2 Weighted-CEL0 Minimization

We now test the effectiveness of S-FISTA when used as inner solver for computing solutions of the non-convex sparse optimization problem (8) via the inexact IRL1 Algorithm 2. For the following experiments, we consider the ISBI SMLM 2013 data, composed of $K = 361$ images representing 8 tubes of 30 nm diameter, see Fig. 2(b) for an exemplar frame and Fig. 2(c) for the average (wide-field) image $\bar{\mathbf{y}} = \frac{1}{K}\sum_{i=1}^{K}\mathbf{y}_i$ where each $\mathbf{y}_i$, $i = 1, \ldots, K$ is a LR image $\mathbf{y}_i \in \mathbb{R}^M_{>0}$ with $M = 64^2$ and SR factor $L = 4$. The Gaussian PSF $\mathbf{h}$ is fully-defined in terms of its Full Width at Half Maximum which for these simulations is FWHM $= 258.2$ nm, and the noise is signal-dependent, so the choice of a Poisson-type fidelity appears natural, see [11].



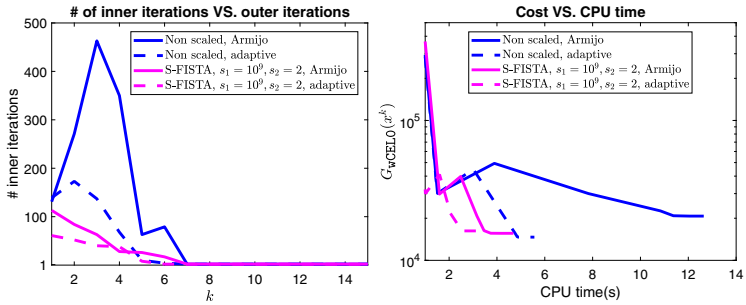
(a) Relative rates VS it. (b) Relative rates VS time (c) L_k estimates.

Fig. 3. FISTA VS. S-FISTA with Armijo (top row) and adaptive (bottom row) backtracking with $L_0 = 100$ for different choices of s_1 and s_2 for problem (14) on `density` image.

Due to the large number of data in the image stack, the use of fast optimization solvers is crucial in these applications to compute solutions of (8) in limited time. Note that for CEL0, in [9], a standard FISTA routine with fixed time-step $\tau = 1/\mathcal{L}_{\texttt{CEL0}}$ with $\mathcal{L}_{\texttt{CEL0}} = L^2 \max(\mathcal{F}(\mathbf{h}))^2$ was used. However, in the case of weighted-ℓ_2 fidelity, the estimation (12) may possibly lead to very pessimistic estimates. As a consequence, the convergence speed of the inner FISTA routine and of the outer IRL1 algorithm may suffer. Motivated by the previous example in Sect. 4.1, we thus tested Algorithm 2 endowed with standard (non-scaled) FISTA and S-FISTA (for selected parameters scaling parameters $s_1 = 10^9, s_2 = 2$ in (13)) as inner routines and compare their computational efficiency when used to solve (8) on the ISBI 2013 test image in Fig. 2b.

The parameters are set as follows: $\lambda = 6.25$, $\mathbf{x}^0 = \mathbf{A}^T\mathbf{y}$, maximum number of IRL1 iterations `maxIt_IRL1` $= 15$, maximum number of inner S-FISTA iterations `maxIt_SFISTA` $= 1000$, $L_0 = 1/\tau_0 = 0.01$, maximum number of backtracking iterations `maxIt_bt` $= 10$, $\rho = 0.9$, $\delta \in \{0.99, 1\}$ and $t_0 = 1.01$. As a stopping criterion for the inner S-FISTA solver, used a control on the relative cost (11) for between two successive inner iterates with tolerance of `tol` $= 10^{-7}$.

Our computational results are reported in Fig. 4. In particular, Fig. 4(a) shows that differently from non-scaled and non-adaptive FISTA which in the early iterations of IRL1 requires a large number of inner iterations to reach convergence, the use of scaled and adaptive algorithms significantly reduces this drawback. The same behavior is clearly shown in Fig. 4(b) where the decreasing of the non-convex $G_{\texttt{wCEL0}}$ cost is shown as a function of the computational time. In Fig. 2(d) the reconstruction obtained with the scaled and adaptive inner routine is shown and in Table 1 a comparison of the quantitative assessment of



(a) Inner VS. outer IRL1 iterations (b) Cost $G_{\mathtt{wCEL0}}$ VS. CPU times

Fig. 4. Computational improvements of using IRL1 Algorithm 2 combined with non-scaled FISTA and S-FISTA Algorithm 1. Left: number of inner iterations for convergence of the inner routine VS. outer iteration number $k \geq 1$. Right: decrease of cost functional $G_{\mathtt{wCEL0}}$ along CPU times.

Table 1. Jaccard index $\mathbf{J}_\beta := \frac{\mathrm{CD}}{\mathrm{CD+FN+FP}} \in [0,1]$, with CD correct detections, FN false negatives and FP false positives up to tolerance radius of $\beta \in \mathbb{N}$ pixels. Values obtained as average over the frames.

	J_0	J_2	J_4		J_0	J_2	J_4
FISTA Armijo	0.0379	0.3108	0.3515	S-FISTA Armijo	0.0264	0.3401	0.4811
FISTA adapt.	0.0344	0.3071	0.3442	S-FISTA adapt.	0.0233	0.2901	0.3785

localisation precision in terms of Jaccard index is reported. We observe that the use of adaptive S-FISTA favors a much faster decreasing of the cost without undermining the localisation precision, making its use amenable for the time-consuming processing of the entire image stack.

5 Conclusions

We considered a scaled and adaptive variant of the well-known FISTA algorithm for the efficient numerical solution of composite convex optimization problems. The combination of a variable metric scaling with the adaptive adjustment of the algorithmic step-size has recently been shown in [14] to favor faster convergence in function values for problems with signal-dependent (Poisson-type) data terms. We applied the proposed S-FISTA to two image super-resolution problems where sparsity is promoted either via standard ℓ_1 penalty or via an exact continuous relaxation of the ℓ_0 norm (see [11,17]). Our results show that S-FISTA significantly improves upon the computational performance of standard FISTA when used both in convex scenarios and as inner solver for non-convex iteratively reweighted ℓ_1 schemes.

Acknowledgments. LC and SR acknowledge the support of the IEA CNRS grant VaMOS. LC acknowledges the support of the EU H2020 RISE project NoMADS, GA

777826. All authors acknowledge the support of the Italian INdAM group on scientific calculus GNCS.

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